

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250268384

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

YANG; Kang et al.

ARM REST SUPPORT WITH ADJUSTABLE HEIGHT

Abstract

An arm rest support with adjustable height, including a support body and adjustment members connected to two sides of the support body respectively; the adjustment members are movable with respect to the support body to achieve different supporting heights of the support body. A cross section of each adjustment member has different length and width. When the adjustment members are moved such that a surface of each adjustment member corresponding to the length of the cross section thereof is placed flat on a supporting surface, an additional supporting height is achieved. When the adjustment members are moved such that a surface of each adjustment member corresponding to the width of the cross section thereof is placed flat on the supporting surface, a further additional supporting height is achieved. Thus, two additional supporting heights are achieved for the arm rest support to meet the demands of different users.

Inventors: YANG; Kang (Huizhou, CN), YANG; Jingang (Sichuan, CN)

Applicant: YANG; Kang (Huizhou, CN); YANG; Jingang (Sichuan, CN)

Family ID: 1000007899188

Appl. No.: 18/666853

Filed: May 17, 2024

Foreign Application Priority Data

CN 2024203737625

Feb. 28, 2024

Publication Classification

Int. Cl.: A47C7/54 (20060101)

U.S. Cl.:

CPC A47C7/541 (20180801); A47C7/546 (20130101);

Background/Summary

BACKGROUND OF THE INVENTION

[0001] The present invention relates to an arm rest support with adjustable height.

[0002] Nowadays, the market provides various arm rest supports to support the hands and arms of a user, so that dangling arms can be relieved from muscle tension. However, existing arm rest supports do not allow supporting height adjustments, thereby failing to meet the demands of different users.

BRIEF SUMMARY OF THE INVENTION

[0003] In view of the aforesaid disadvantages now present in the prior art, the present invention provides an arm rest support with adjustable height, achieved through the following technical proposal:

[0004] An arm rest support with adjustable height, comprising a support body; adjustment members are connected to two sides of the support body respectively; the adjustment members are movable with respect to the support body to achieve different supporting heights of the support body.

[0005] Further, the adjustment members are rectangular prisms each having a cross section in which a length and a width thereof are not identical.

[0006] Further, the adjustment members are pivotally connected to the support body to enable the adjustment members to move pivotally with respect to the support body.

[0007] Further, the support body is provided with an opening.

[0008] Further, the arm rest support further comprises a filler block; the filler block removably fills in the opening, and the filler block is movable relative to the support body so as to stack on the support body to increase a supporting height of the arm rest support.

[0009] Further, the filler block is provided with a storage groove.

[0010] The present invention has the following beneficial effects: In the present invention, a cross section of each of the adjustment members has a length and a width which are not identical. When the adjustment members are moved relative to the support body to a first position where a surface of each of the adjustment members corresponding to the length of the cross section of each of the adjustment members is placed flat on a supporting surface, an additional supporting height is achieved. When the adjustment members are moved relative to the support body to a second position where a surface of each of the adjustment members corresponding to the width of the cross section of each of the adjustment members is placed flat on the supporting surface, a further additional supporting height is achieved. Thus, movements of the adjustment members relative to the support body can provide two additional supporting heights for the support body. Accordingly, the supporting height of the arm rest support can be adjusted to meet the demands of different users.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 illustrates a structure of the arm rest support of the present invention.

[0012] FIG. 2 illustrates a structure of the arm rest support in another supporting height according to the present invention.

[0013] FIG. 3 illustrates a structure of the arm rest support in still another supporting height according to the present invention.

[0014] FIG. 4 is a schematic illustration showing the arm rest support of the present invention in a state of use.

[0015] FIG. 5 is an exploded structural view of the arm rest support containing a filler block according to the present invention.

[0016] FIG. 6 is an assembled view of the arm rest support and the filler block.

[0017] FIG. 7 is another structural view of the arm rest support of the present invention having a yet further supporting height.

DETAILED DESCRIPTION OF THE INVENTION

[0018] The technical solution of the present invention will be described in a clear and complete manner as follows with reference to the drawings of some exemplary embodiments of the present invention. It is apparent that the described embodiments are only some but not all of the embodiments of the present invention. Based on the embodiments of the present invention, all other embodiments obtainable by those skilled in the art without exercising inventive effects shall fall within the scope of protection of the present invention.

[0019] As shown in FIG. 1, the present invention provides an arm rest support with adjustable height, comprising a support body 1; adjustment members 2 are connected to two sides of the support body 1 respectively; the adjustment members 2 are movable with respect to the support body 1 to achieve different supporting heights of the support body 1.

[0020] In the present invention, a cross section of each of the adjustment members 2 has a length and a width which are not identical. When the adjustment members are moved relative to the support body 1 to a first position where a surface of each of the adjustment members 2 corresponding to the length of the cross section of each of the adjustment members is placed flat on a supporting surface, an additional supporting height is achieved (as shown in FIG. 2). When the adjustment members are moved relative to the support body 1 to a second position where a surface of each of the adjustment members 2 corresponding to the width of the cross section of each of the adjustment members is placed flat on the supporting surface, a further additional supporting height is achieved (as shown in FIG. 3). Thus, movements of the adjustment members 2 relative to the support body 1 can provide two additional supporting heights for the support body 1. Accordingly, the supporting height of the arm rest support can be adjusted to meet the demands of different users. FIG. 4 illustrates a state of use of the arm rest support of FIG. 3.

[0021] In a further aspect of the present invention, as shown in FIGS. 2 and 3, the adjustment members 2 are rectangular prisms each having a cross section in which a length and a width thereof are not identical. By moving the adjustment members to different positions relative to the support body so that the surface of each of the adjustment members corresponding to the length of the cross section of each of the adjustment members or the surface of each of the adjustment members corresponding to the width of the cross section of each of the adjustment members can be selectively placed flat on the supporting surface to provide two additional supporting heights.

[0022] In another aspect of the present invention, as shown in FIGS. 2 and 3, the adjustment members 2 are pivotally connected to the support body 1 to enable the adjustment members 2 to move pivotally with respect to the support body 1.

[0023] In a further aspect of the present invention, as shown in FIG. 5, the support body 1 is provided with an opening 11. When a user is seated, the opening 11 is configured to at least partially accommodate the user's body as the support body 1 is placed on the user's body. The opening 11 allows for stable placement of the support body 1, and the opening 11 also provides a space for potential adjustment of a relative position between the support body 1 and the user's body in order to meet the needs of different users.

[0024] In another aspect of the present invention, as shown in FIG. 5, the arm rest support further comprises a filler block 3. The filler block 3 is used to fill the opening 11 and is movable (including but not limited to "pivotable") relative to the support body 1 so as to stack on the support body 1 to further increase the supporting height of the arm rest support. When the filler block is filled into the opening, sides of the filler block flush with corresponding sides of the opening so that surfaces of the filler block also flush with corresponding surfaces of the arm rest support to render an integral

appearance of the arm rest support, which can be used as a seat cushion (as shown in FIG. 6). [0025] In a further aspect of the present invention, as shown in FIG. 7, the filler block 3 is provided with a storage groove 31. The storage groove 31 can be used to store items. [0026] It should be noted that, new embodiments can be obtained by combining the different embodiments and the features disclosed therein, provided that such combination does not result in any contradiction. [0027] The above description represents preferred embodiments of the present invention and should not be construed as imposing any limitation on the present invention. The scope of protection of the present invention should be determined by the claims. Although the present invention has been disclosed with the preferred embodiments mentioned above, they are not intended to limit the present invention. Any other embodiments with equivalent technical effects through alternative configurations based on slight modifications or variations of the present invention made by those skilled in the art within the technical scope of the present invention and in accordance with the teachings of the present invention as disclosed above, are still within the scope of the present invention.

Claims

1. An arm rest support with adjustable height, comprising a support body; adjustment members are connected to two sides of the support body respectively; the adjustment members are movable with respect to the support body to achieve different supporting heights of the support body.
 2. The arm rest support with adjustable height of claim 1, wherein the adjustment members are rectangular prisms each having a cross section in which a length and a width thereof are not identical.
 3. The arm rest support with adjustable height of claim 1, wherein the adjustment members are pivotally connected to the support body to enable the adjustment members to move pivotally with respect to the support body.
 4. The arm rest support with adjustable height of claim 1, wherein the support body is provided with an opening.
 5. The arm rest support with adjustable height of claim 4, wherein the arm rest support further comprises a filler block; the filler block removably fills in the opening, and the filler block is movable relative to the support body so as to stack on the support body to increase a supporting height of the arm rest support.
 6. The arm rest support with adjustable height of claim 5, wherein the filler block is provided with a storage groove.
-